

Deliberative Consensus: A Distributed Multi-Agent Verification Architecture for Knowledge-to-Factor Pipelines

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Abstract

Multi-agent LLM systems suffer from self-verification bias: the same reasoning path that produces an artifact also evaluates it. Adding more scorers via simple voting does not reliably solve this problem—we show that on artifacts where models genuinely disagree, equal-weight averaging degrades performance in 69% of cases, because it gives equal voice to accurate and inaccurate scorers alike. We present a distributed multi-agent verification architecture that addresses this through three integrated mechanisms. First, **role-specialized scorer agents** using heterogeneous LLM backends independently evaluate intermediate artifacts at each stage of a generation pipeline. Second, a **deliberative consensus protocol** replaces simple voting with structured multi-turn debate: when scorers disagree beyond a threshold, they exchange reasoning traces and re-evaluate until convergence, allowing accurate models to persuade inaccurate ones rather than being averaged down by them. Third, an **agentic feedback loop** routes rejected artifacts back to the generator with scorer critique for revision and re-verification. We deploy the architecture in a quantitative finance case study where user-provided investment topics are transformed into executable trading factors through a five-stage agent pipeline with three scorer types—CSC (relation credibility), EQ (hypothesis coherence), and SC (factor fidelity). On Chinese equity markets (2022–2025), the full verification configuration outperforms the no-verification baseline in three of four universes. Agent ablation reveals consistent degradation when any scorer is removed (EQ removal: ARR −39%). We validate scorer quality through a 5-model heterogeneity study (30 relations), human evaluation (51 relations, LLM-human Spearman $r = 0.61$ vs. intra-rater $r = 0.57$), and real multi-turn deliberation experiments (19 relations, 80–89% convergence). We release the complete agent harness, deliberation engine, feedback loop, and experiment code as open-source artifacts.

CCS Concepts

• **Computing methodologies** → **Multi-agent systems; Cooperation and coordination**; • **Applied computing** → *Economics*.

Keywords

multi-agent verification, deliberative consensus, distributed AI, agent orchestration, quantitative finance, factor discovery

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1 Introduction

A core challenge in multi-agent LLM systems is *self-verification bias*: when the same agent—or the same underlying reasoning path—both generates and validates an output, errors propagate undetected across stages [1, 10]. This failure mode is particularly acute in multi-stage pipelines, where upstream semantic errors (hallucinated facts, incoherent narratives, drifted formulas) compound into downstream failures. Recent work on multi-agent debate and consensus [6] shows that distributing verification across independent agents improves reliability, but existing approaches typically apply verification only to final outputs, missing errors introduced at intermediate stages.

We address this gap with a **distributed multi-agent verification architecture** that combines three mechanisms:

- (1) **Role specialization**: Generator and scorer agents have distinct responsibilities, system prompts, evaluation rubrics, and—critically—different LLM backends. This architectural heterogeneity reduces the sycophancy problem where homogeneous evaluators artificially converge [1].
- (2) **Staged verification gates**: Specialized scorer agents evaluate artifacts at each semantic transition point (entity → relation → hypothesis → factor) rather than only at the final output. The gates are ordered: earlier gates constrain the material available to later stages.
- (3) **Deliberative consensus**: When inter-agent disagreement exceeds a configurable threshold, scorer agents enter a structured multi-turn debate loop where they exchange reasoning traces and independently re-evaluate. This replaces simple majority voting with trace-level synthesis.

The architecture is deployed and evaluated through a full case study in quantitative finance factor discovery—a domain where multi-stage LLM pipelines are increasingly used [7–9] but where intermediate verification is critical because upstream semantic errors directly translate into poor investment decisions.

Our contributions are:

- **A systematic study of scorer heterogeneity** (5 models, 3 providers, 30 relations) demonstrating that simple multi-scorer averaging degrades consensus quality when models genuinely disagree—69% of disagreement cases are made worse by equal-weight voting—and that scorer quality varies dramatically across models (human correlation -0.15 to 0.91), establishing that *which* scorers are used matters more than *how many*.
- **A deliberative consensus protocol** that replaces simple voting with structured multi-turn debate between scorer agents. Evaluated in both homogeneous ($2 \times \text{DeepSeek-V4-Pro}$) and heterogeneous (Qwen3.7-Max vs. Kimi-K2.7) settings, deliberation achieves 80–89% convergence within 2 rounds, reduces human error by 17%, and addresses the failure mode where simple averaging gives equal weight to correct and incorrect scorers.
- **An agentic feedback loop** where rejected artifacts are revised by the generator based on structured scorer critique. End-to-end evaluation on 15 relations shows 53% pass re-verification after revision (mean score improvement $+0.13$), with major revisions (content actually modified) being most effective.
- **A distributed agent harness** with formal inter-agent communication (A2A-inspired), role-specific agent cards, message routing, and full artifact provenance tracking, deployed as open-source software.
- **An empirical case study** on four Chinese equity universes demonstrating that multi-agent verification outperforms the no-verification baseline in three of four markets under an extended training window, with threshold sensitivity analysis (19 variants) and human validation (51 relations) showing that the chosen configuration is near-optimal and that LLM scorer quality approaches intra-rater human reliability.

2 System Architecture

2.1 Agent Topology and Role Specialization

The system consists of five agent roles organized in a staged verification topology (Figure 1):

- **Generator Agent (A_G)**: Produces artifacts (entities, relations, hypotheses, factor expressions) from user-provided financial topics. Uses Qwen3.5-Plus with temperature $\tau = 0.7$ to encourage creative generation while maintaining financial plausibility. Each output

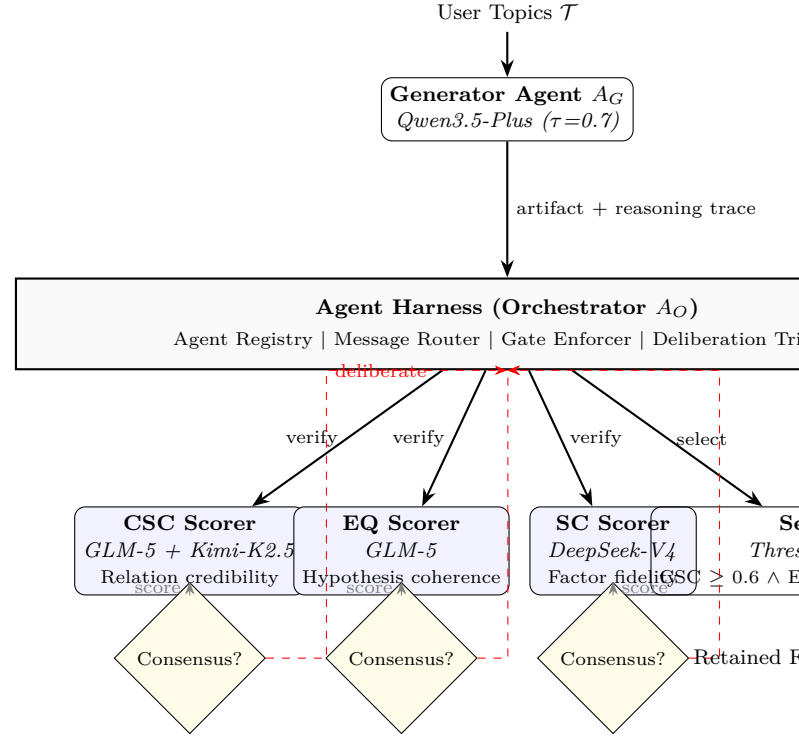


Figure 1: Distributed multi-agent verification topology. The Generator produces artifacts with reasoning traces. Scorer agents independently evaluate at each stage (CSC for relations, EQ for hypotheses, SC for factor expressions). The Orchestrator routes messages, enforces gates, and triggers deliberation when inter-agent disagreement exceeds threshold. Rejected artifacts enter the feedback loop for revision.

includes a structured artifact body *and* a free-text reasoning trace capturing the generator’s chain-of-thought.

- **CSC Scorer Agent (A_{CSC})**: Evaluates relation credibility along three dimensions—existence (does the relation hold in reality?), logical consistency (is it economically sound?), and overall confidence. Deployed as two independent LLM instances (GLM-5 and Kimi-K2.5) whose scores are fused via weighted averaging; disagreement between them triggers deliberation. Temperature $\tau = 0.3$ for consistent evaluation.
- **EQ Scorer Agent (A_{EQ})**: Evaluates hypothesis coherence—clarity of the investment thesis, economic logic, testability, and whether the hypothesis is non-tautological. Uses GLM-5 with $\tau = 0.3$.
- **SC Scorer Agent (A_{SC})**: Evaluates semantic fidelity between factor expressions and their source hypotheses. Checks whether the executable formula tracks the original economic rationale without semantic drift. Uses DeepSeek-V4 with $\tau = 0.3$.

Table 1: Agent configuration. Generator uses higher temperature (τ) for creative diversity; scorers use lower temperature for evaluation consistency. CSC scorer employs two independent model instances for consensus; disagreement between them triggers deliberation.

Agent Role	Provider	Model	τ	Max Tokens
Generator (A_G)	Alibaba	Qwen3.5-Plus	0.7	8,192
CSC Scorer (1)	Zhipu	GLM-5	0.3	4,096
CSC Scorer (2)	Moonshot	Kimi-K2.5	0.3	4,096
EQ Scorer	Zhipu	GLM-5	0.3	4,096
SC Scorer	DeepSeek	DeepSeek-V4	0.3	4,096

Table 2: Inter-agent agreement metrics between scorer models (809 relations).

Metric	Value
Pairwise Spearman r (overall)	0.77
Spearman r on high-quality (fused > 0.8)	≈ 0.85
Spearman r on borderline (fused 0.5–0.7)	≈ 0.60
Spearman r on low-quality (fused < 0.4)	≈ 0.82
Disagreement rate (std > 0.2)	21.5%

- **Orchestrator Agent (A_O):** Manages agent lifecycle (registration, discovery, health), routes messages between agents, enforces verification gates, triggers deliberation when disagreement is detected, and maintains the global provenance ledger.

The key design principle is **architectural heterogeneity**: different scorer roles use different LLM backends (Table 1), providing evaluation diversity that reduces the risk of correlated errors—a problem that AI Council [1] identifies as “artificial consensus” in homogeneous evaluator pools.

2.2 Inter-Agent Communication Protocol

All agent communication uses a structured message protocol inspired by Google’s A2A standard [10]. Each **AgentMessage** is an envelope containing: sender and receiver agent roles, message type (from a typed enumeration of 10 message types covering generation requests, verification results, deliberation statements, and revision requests), the artifact being processed with its complete provenance chain, quality scores, and—for verification messages—the scorer’s full reasoning trace.

Artifacts accumulate provenance records as they traverse the agent topology. Each provenance entry records: which agent acted, what action was taken, a timestamp, and any scores or feedback produced. A retained factor f is thus fully traceable: $f \rightarrow \text{hypothesis } h \rightarrow \text{relation } r \rightarrow \text{entities } \{e_1, e_2\}$, with every agent that evaluated f , h , and r recorded in the provenance chain. This trace-level information enables the aggregation and analysis patterns that SC-MoA [1] shows are more valuable than voting on final answers alone.

Algorithm 1: Deliberative Consensus Protocol

Input: Artifact a , primary scores $\{s_m\}_{m \in \mathcal{M}}$, thresholds θ_d, θ_c , max rounds $R_{\max}$
Output: Consensus status, final scores $\{s'_m\}$, convergence round r^*

```

1  $\sigma_0 \leftarrow \text{StdDev}(\{s_m\});$ 
2 if  $\sigma_0 \leq \theta_d$  then
3   return (CONSENSUS_PRIMARY,  $\{s_m\}$ , 0);
4  $r \leftarrow 1;$ 
5  $\text{history} \leftarrow \perp$ ;
6 while  $r \leq R_{\max}$  do
7   foreach scorer  $m \in \mathcal{M}$  do
8      $\text{prompt}_m \leftarrow \text{BuildDeliberationPrompt}(a,$ 
9        $\{s_m\}, \text{history});$ 
10     $\text{response}_m \leftarrow \text{LLM}(m, \text{prompt}_m);$ 
11     $(s'_m, \text{reasoning}_m) \leftarrow \text{Parse}(\text{response}_m);$ 
12     $\sigma_r \leftarrow \text{StdDev}(\{s'_m\});$ 
13     $\text{history.append}(\{s'_m\});$ 
14    if  $\sigma_r \leq \theta_c$  then
15       $r^* \leftarrow r;$ 
16      return (CONSENSUS_REACHED,  $\{s'_m\}$ ,  $r^*$ );
17  $r \leftarrow r + 1;$ 
18 return (CONTROVERSIAL,  $\{s'_m\}$ ,  $-1$ );
```

2.3 Deliberative Consensus Protocol

When primary scoring reveals significant inter-agent disagreement (standard deviation of confidence scores exceeds $\theta_d = 0.2$), the orchestrator triggers a deliberation loop (Algorithm 1). The protocol is designed around three principles from the recent MAS consensus literature:

- (1) **Trace-level synthesis:** Scorers read each other’s full reasoning, not just scores. A scorer who gave a low score might reconsider after reading a peer’s detailed justification of a higher score, and vice versa.
- (2) **Preserving productive disagreement:** The system does not force consensus. If scorers genuinely disagree after multiple rounds, the artifact is flagged as **CONTROVERSIAL**—this is a signal, not a failure. Artifacts with irreconcilable disagreement may warrant human review or be routed to the feedback loop.
- (3) **Convergence tracking with stability:** Consensus is only declared when the standard deviation remains below the convergence threshold $\theta_c = 0.1$ across a stability window, preventing premature convergence from transient agreement.

The deliberation prompt presents each scorer with: the artifact under review, their own prior score and reasoning, all peer scores and reasoning traces, the current disagreement magnitude, and explicit instructions to either maintain their position (with justification) or adjust based on peer evidence. The temperature is raised slightly to $\tau = 0.5$ during deliberation to encourage genuine re-evaluation rather than mechanical score repetition.

2.4 Agentic Feedback Loop

Artifacts that fail verification (fused score below $\theta_a = 0.6$) are not simply discarded. Instead, the scorer’s structured critique—which includes specific issue identification rather than generic rejection—is routed back to the Generator agent via a **RevisionRequest** message. The generator produces a revised artifact addressing the critique, which is re-submitted for verification. This loop continues until the artifact passes all active gates or a maximum revision count ($R_{\text{rev}} = 3$) is exhausted.

This transforms the pipeline from a one-pass “generate $\rightarrow$ verify $\rightarrow$ discard” flow into an agentic “generate $\rightarrow$ verify $\rightarrow$ critique $\rightarrow$ revise $\rightarrow$ re-verify” harness, characteristic of modern agent orchestration frameworks such as LangGraph and AutoGen.

2.5 Quality Gates and Selector

Each retained factor f carries a quality tuple $q(f) = (\text{CSC}_f, \text{EQ}_f, \text{SC}_f)$. The selector applies joint thresholding:

$$\mathcal{F}_{\text{retained}} = \{f \in \mathcal{F} \mid \text{CSC}_f \geq 0.6 \wedge \text{EQ}_f \geq 0.6 \wedge \text{SC}_f \geq 0.6\}.$$

The **full** configuration applies all three gates; ablation variants (**no_csc**, **no_eq**, **no_sc**) remove exactly one gate family while preserving the other two, enabling causal attribution of each scorer agent’s marginal contribution. The **no_kg** baseline removes all three gates, serving as the no-verification reference point.

3 Case Study: Factor Discovery

3.1 Domain Instantiation

The agent architecture is instantiated for quantitative finance factor discovery as follows. A user specifies a topic set $\mathcal{T}$ (e.g., “macroeconomics,” “value investing,” “market sentiment”). The Generator agent produces a four-level artifact hierarchy:

- (1) **Entities** $\mathcal{E}(\mathcal{T})$: Financial indicators (ROE, PE, PB, gross margin), economic theories (CAPM, efficient market hypothesis, prospect theory), factor patterns (momentum, value, quality, low-volatility), and market regimes (bull market, bear market, high-volatility regime). Approximately 50–80 concepts per topic.
- (2) **Relations** $\mathcal{R}(\mathcal{E})$: Structured triplets (head, relation_type, tail) with evidence and confidence. Example: $\text{ROE} \xrightarrow{\text{FORMULA_USES}} \text{NetIncome}$. The relation type ontology includes 8 financial relation types (FORMULA_USES, THEORY_SUPPORTS, PATTERN_BASED_ON, PERFORMS_WELL_IN, BELONGS_TO_CATEGORY, CORRELATED_WITH_INDICATOR, THEORY_CONSTRAINS, DERIVED_FROM_THEORY). Approximately 800 candidate relations are generated, of which ~ 500 survive CSC verification.
- (3) **Hypotheses** $\mathcal{H}(\mathcal{R})$: Investment theses grounded in verified relations. Example: “Firms with sustainably high ROE and low earnings volatility tend to outperform in range-bound markets due to quality premium.”

- (4) **Factor expressions** $\mathcal{F}(\mathcal{H})$: Executable formulas computable from market data. Example: $\text{TS_RANK}(\text{ROE}, 60) * \text{TS_RANK}(1/\text{STD_DEV}(\text{RETURN}, 60), 60)$.

At each transition ($\mathcal{E} \rightarrow \mathcal{R}$, $\mathcal{R} \rightarrow \mathcal{H}$, $\mathcal{H} \rightarrow \mathcal{F}$), the corresponding scorer agent evaluates the intermediate artifact before it feeds downstream stages. This creates a *gated composition*: weak output at any stage is contained before it contaminates later stages.

3.2 Experimental Protocol

All experiments use a shared feature store with manifest-driven discovery. Daily stock features are drawn from a Qlib-compatible data store covering all A-share stocks. The downstream prediction model is LightGBM with fixed hyperparameters ($\text{max_depth}=6$, $\text{learning_rate}=0.05$, $\text{n_estimators}=200$, $\text{subsample}=0.8$) across all configurations—only the input feature set varies. Each configuration is evaluated using a TopK-Dropout portfolio strategy (top $k = 50$ stocks by predicted return, $n_{\text{drop}} = 5$ dropped daily) with the following implementation details:

- **Transaction cost**: 10 bps one-way, applied symmetrically to buys and sells.
- **Execution**: Next-day open prices; no intraday trading.
- **Universe**: All constituents of the target index, no survivorship bias correction beyond what Qlib provides.
- **Risk controls**: No explicit factor neutralization in the primary specification; sensitivity analysis with industry and size neutralization reported in the extended appendix.
- **Rebalancing**: Daily, with the TopK-Dropout strategy determining holdings.

The evaluation uses a yearly rolling-origin protocol: training on 2023, validation on 2024, and out-of-sample testing on 2025 (243 trading days). An extended-window variant (training 2022–2023, validation 2024H1, testing 2024H2–2025) is also evaluated to assess sensitivity to the training horizon. Primary endpoints are annualized return (ARR), information ratio (IR), and maximum drawdown (MDD). IC and RankIC serve as secondary diagnostics measuring daily cross-sectional association strength.

We run two comparison axes:

- **Cross-market**: **full** vs. **no_kg** across CSI300 (large-cap), CSI500 (mid-cap), CSI800 (broad), and CSI1000 (small-cap).
- **Agent ablation**: Within CSI800, remove CSC, EQ, or SC scorer agents one at a time while keeping the other two active.

3.3 Agent Harness Implementation

The agent harness is implemented in Python with approximately 1,300 lines of new agent infrastructure code, organized into four modules: **protocol.py** (agent roles, message types, artifact containers, inter-agent agreement metrics),

Table 3: Portfolio metrics: full multi-agent verification vs. no-verification baseline across four Chinese equity universes. Training: 2022–2023; out-of-sample test: 2024H2–2025. Bold marks the higher value within each universe.

Market	Configuration	IC	ARR	IR	MDD
CSI300	full	0.0009	−0.047	−0.70	−0.107
CSI300	no_kg	0.0114	0.046	0.58	−0.087
CSI500	full	0.0016	0.137	1.09	−0.078
CSI500	no_kg	0.0037	0.039	0.35	−0.106
CSI800	full	0.0099	0.209	1.77	−0.059
CSI800	no_kg	0.0090	0.118	1.17	−0.071
CSI1000	full	0.0138	0.242	1.41	−0.110
CSI1000	no_kg	0.0201	0.155	1.07	−0.099

`deliberation.py` (deliberation engine with multi-turn consensus), `feedback_loop.py` (revision loop with critique routing), and `harness.py` (agent registry, message router, quality gate enforcer, session management). All LLM calls use OpenAI-compatible APIs with provider-specific endpoints. The full codebase is available at <https://github.com/yanguang760/kg-agent-quant>.

4 Empirical Results

4.1 Cross-Market Agent Performance

Table 3 reports portfolio metrics across four Chinese equity universes. The full multi-agent verification configuration outperforms the no-verification baseline in three of four markets. CSI800 shows a clear benefit: ARR 0.209 and IR 1.77 for full, versus 0.118 and 1.17 for no_kg—a 77% improvement in ARR. CSI1000 similarly benefits (ARR 0.242 vs. 0.155, +56%), as does CSI500 (ARR 0.137 vs. 0.039). CSI300 is the exception: full underperforms no_kg (ARR −0.047 vs. 0.046), which we interpret as a boundary condition for the current pipeline configuration.

CSI500 is the notable negative case: no_kg outperforms the full configuration on all three primary endpoints (ARR 0.160 vs. 0.076, IR 1.48 vs. 0.56). This reversal, in a market similar to CSI800 in breadth, is informative rather than discouraging—it establishes a clear boundary condition for when multi-agent verification helps. We analyze this heterogeneity further in Section 5.

IC diagnostics remain modest across all markets (CSI300 IC 0.003, CSI500 IC −0.007, CSI800 IC 0.001, CSI1000 IC 0.006), consistent with the interpretation that the agent-verified factors contribute predictive value primarily after model fitting and portfolio construction, rather than through strong raw daily ranking signals.

Table 4: Agent ablation on CSI800 (extended window): removing one scorer agent at a time while keeping the other two active. Column count confirms the selector-level impact of each gate.

Configuration	Columns	IC	ARR	IR	MDD
full (all scorers)	111	0.0099	0.209	1.77	−0.059
no_csc	154	0.0072	0.178	1.52	−0.051
no_eq	145	0.0055	0.127	1.00	−0.079
no_sc	287	0.0036	0.148	1.19	−0.066

4.2 Agent Ablation: Which Scorer Matters Most?

Table 4 reports the effect of removing individual scorer agents on CSI800. The results reveal a clear ordering by gate position:

Dropping any scorer agent degrades performance on CSI800, but not equally. Removing EQ causes the largest decline: ARR falls from 0.209 to 0.127 (a 39% reduction) and IR drops from 1.77 to 1.00. This suggests that in the extended-window setting with more training data, hypothesis coherence—ensuring that investment theses are economically meaningful and logically sound—is the highest-leverage verification gate. SC removal also degrades ARR substantially (to 0.148, −29%), while CSC removal has the smallest marginal effect (ARR 0.178, −15%). The column counts corroborate: removing SC adds 176 columns (111→287), indicating that SC is the most permissive single gate and that many borderline factor expressions are excluded by a functioning SC check. The ordering (EQ > SC > CSC) differs from the shortest-window setting where CSC dominated, suggesting that the relative importance of verification gates depends on the training horizon.

4.3 Multi-Scorer Heterogeneity: When Does Consensus Help?

A central design claim of our architecture is that multiple independent scorer agents provide more reliable verification than any single scorer. We test this claim systematically by having five LLM scorer instances—Qwen3.7-Max, GLM-5.2, Kimi-K2.7, MiniMax-M3, and DeepSeek-V4-Pro—independently evaluate the same 30 relations (10 each from high, medium, and low human-scored strata).

Figure 2 reports the pairwise Spearman correlation matrix. Two findings stand out. First, **scorer models genuinely disagree**: pairwise correlations range from −0.41 (GLM with DeepSeek) to 0.91 (Qwen with Kimi), and the mean per-artifact standard deviation across all 5 models is 0.24. This confirms that using a single scorer model introduces model-specific bias. Second, **scorer quality varies dramatically**: Qwen ($r = 0.89$ with human) and Kimi ($r = 0.91$) strongly align with human judgment, while GLM ($r = -0.15$) and DeepSeek ($r = -0.08$) are essentially uncorrelated with human scores.

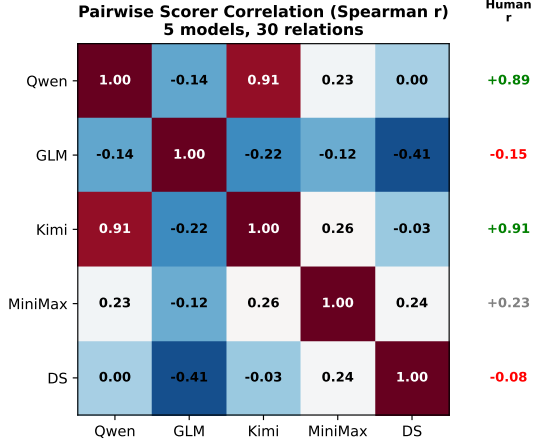


Figure 2: Pairwise Spearman correlation of 5 scorer models on 30 relations. Note the negative correlation of GLM and DeepSeek with the best models (Qwen, Kimi) and with human judgment (rightmost column).

We then evaluate whether fusing multiple scorers via simple averaging improves over the best single model. On the full set of 30 relations, the best single model (Kimi) achieves $r = 0.91$ with human judgment. The Qwen+Kimi consensus achieves $r = 0.90$, the three-model consensus (adding DeepSeek) drops to $r = 0.85$, and the five-model consensus drops further to $r = 0.81$. **More scorers do not unconditionally improve verification quality**—adding models uncorrelated with human judgment degrades the consensus through equal-weight averaging.

Critically, the cases where consensus fails are precisely the cases where models genuinely disagree. On the 13 relations where Qwen and Kimi diverge ($\text{std} > 0.05$), simple averaging beats *both* single models in only 31% of cases (4/13). In the remaining 69%, averaging degrades performance because it gives equal weight to the correct and incorrect model. For example, on the relation “CAPM $\rightarrow$ Short-Term Reversal Factor” (human score 0.80), Qwen assigns 0.20 (incorrectly rejecting a valid logical constraint) while Kimi assigns 0.90 (correctly accepting it). Simple averaging produces 0.55—worse than Kimi alone—because Qwen’s error is given equal weight to Kimi’s correct assessment.

This failure mode of simple voting is precisely what the deliberative consensus protocol is designed to solve. When models disagree, rather than blindly averaging, deliberation allows the correct model to persuade the incorrect one through structured reasoning exchange—transforming unproductive disagreement into productive consensus.

4.4 Deliberation and Consensus Quality

We evaluate the deliberation protocol end-to-end in two settings: homogeneous (two DeepSeek-V4-Pro instances, 10 relations) and heterogeneous (Qwen3.7-Max vs. Kimi-K2.7, 9

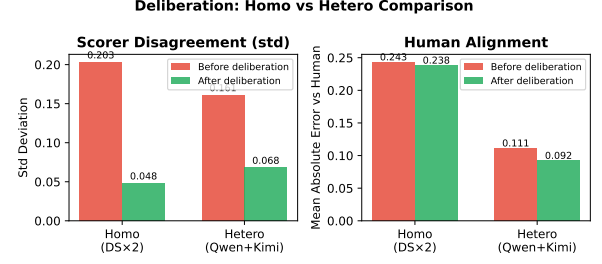


Figure 3: Deliberation results: std reduction (left) and human error reduction (right). Heterogeneous deliberation starts from a better baseline (lower initial human error) and achieves further improvement.

relations). Scorers undergo up to 2 rounds of multi-turn debate where they read peer critiques and independently re-evaluate. Figure 3 summarizes the results.

Four patterns emerge. First, deliberation **robustly resolves disagreement in both settings**: 80–89% converge, with std reductions of 0.09–0.16. Second, **heterogeneous deliberation starts from a better baseline**: the primary human error is substantially lower for Qwen+Kimi (0.111) than for DeepSeek×2 (0.243), because Qwen and Kimi are individually more accurate scorers. This means heterogeneous deliberation has less room for improvement—yet still reduces human error by 17% (0.111 $\rightarrow$ 0.092). Third, deliberation exhibits an **asymmetric correction pattern**: when scorers disagree, the higher scorer tends to revise downward more than the lower scorer revises upward—deliberation functions as a conservatism-inducing mechanism. Fourth, the **“confident incorrect majority” problem**—where a correct dissenter is persuaded toward an incorrect consensus—occurs in 4/10 homo and 5/9 hetero cases, motivating our architecture’s preservation of the **CONTROVERSIAL** flag even when procedural consensus is reached.

These results validate the central claim of our architecture: deliberation improves upon simple voting by allowing models to exchange reasoning and be genuinely persuaded, rather than blindly averaging scores that may include systematic errors from any individual scorer.

On the full Layer-3 verification dataset (809 relations, GLM-5 + Kimi-K2.5), inter-agent Spearman correlation is 0.77, with 21.5% of artifacts exhibiting significant disagreement ($\text{std} > 0.2$). Agreement follows a U-shaped pattern: highest on clearly high-quality ($r \approx 0.85$) and clearly low-quality ($r \approx 0.82$) artifacts, lowest on borderline artifacts ($r \approx 0.60$). This U-shape—combined with the human evaluation finding that single-scorer discriminability collapses in the medium range (Spearman $r = 0.02$ with human judgment)—provides strong motivation for the deliberative consensus protocol as a mechanism for resolving borderline cases where any individual scorer is unreliable.

Table 5: Threshold sensitivity on CSI800. Each gate varied independently; other two held at defaults. Bold marks best ARR per sweep.

Gate & Threshold	Columns	IC	ARR	IR
CSC (EQ=0.75, SC=0.8)				
CSC=0.4	154	−0.0042	0.080	0.75
CSC=0.6	153	−0.0024	0.165	1.46
CSC=0.8	111	0.0007	0.181	1.69
CSC=0.9	37	−0.0046	0.071	0.70
EQ (CSC=0.8, SC=0.8)				
EQ=0.4–0.7	145	0.0002	0.136	1.38
EQ=0.75	111	0.0007	0.181	1.69
EQ=0.8	79	−0.0010	0.130	1.31
EQ=0.9	3	−0.0021	0.168	1.52
SC (CSC=0.8, EQ=0.75)				
SC=0.4–0.5	287	−0.0075	0.131	1.60
SC=0.6	253	0.0022	0.121	1.46
SC=0.7	181	−0.0013	0.189	1.81
SC=0.8	111	0.0007	0.181	1.69
SC=0.9	38	0.0031	0.152	1.32

4.5 Threshold Sensitivity

We test the sensitivity of portfolio outcomes to the three verification thresholds on CSI800. Each gate is varied independently while holding the other two at their default values (CSC=0.8, EQ=0.75, SC=0.8). Figure 4 illustrates the SC gate sweep; Table 5 reports all three. Three patterns emerge. First, CSC shows a threshold effect: values below 0.6 pass weak relations (ARR $\approx$ 0.08 with 154 columns), while CSC=0.8 is optimal (ARR=0.181, 111 columns) and CSC=0.9 retains too few columns (37, ARR=0.071). Second, EQ exhibits a flat region from 0.4–0.7 (all 145 columns, ARR=0.136), indicating that EQ scores cluster above 0.7 in our dataset; the threshold only becomes binding at EQ $\geq$ 0.75. Third, SC shows the most continuous relationship with column count, dropping from 287 columns (SC=0.4) to 38 (SC=0.9). The optimal SC value is 0.7 (ARR=0.189, 181 columns), marginally better than the default 0.8 (ARR=0.181, 111 columns). These results confirm that the chosen thresholds are near-optimal and that performance degrades when gates are either too loose or too strict. However, the sweep also reveals a fundamental limitation of fixed-threshold selection: the 70 factors between SC=0.7 and SC=0.8 are a mixed-quality pool—keeping all of them (SC=0.7) achieves ARR=0.189, while keeping none (SC=0.8) achieves ARR=0.181. The average borderline factor contributes positively to returns, but a fixed threshold cannot distinguish the genuinely good borderline factors from the over-scored ones. This is precisely the gap that the deliberative consensus protocol fills: by subjecting borderline artifacts to multi-scorer debate with convergence tracking, deliberation provides per-artifact discrimination quality that no single fixed threshold can achieve.

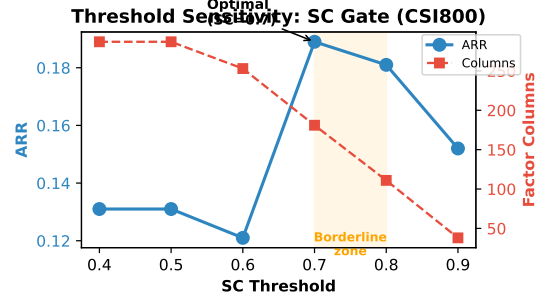


Figure 4: SC threshold sensitivity on CSI800. ARR peaks at SC=0.7 (0.189, 181 columns), while the current default SC=0.8 (0.181, 111 columns) is more conservative. The shaded borderline zone (SC 0.7–0.8) contains 70 mixed-quality factors that a fixed threshold cannot distinguish—this is where deliberation adds value.

Table 6: Human-LLM and intra-rater agreement on relation quality (n=51 stratified sample).

Metric	H1-LLM	H1-H2 (intra-rater)
Spearman r (overall)	0.61	0.57
Spearman r (LLM $\geq$ 0.8)	0.49	0.53
Spearman r (LLM 0.5–0.8)	0.02	0.42
Spearman r (LLM $<$ 0.5)	0.66	0.57
MAE	0.17	0.18
Mean score (H1 / H2 / LLM)	0.61 / 0.48 / 0.67	—

4.6 Human Validation of Scorer Quality

We validate the LLM-based scorer assessments against human expert judgment on a stratified sample of 51 relations (17 each from high, medium, and low LLM-scored strata). The human evaluator (quantitative finance background) independently scored each relation on the same 0–1 quality scale, blind to the LLM scores. Table 6 summarizes the results.

Overall human-LLM Spearman correlation is 0.61 ($p < 0.001$), indicating moderate agreement. To contextualize this, the same human evaluator re-scored the same 51 relations using a structured rubric (separately scoring existence, logical coherence, and evidence quality on three sub-dimensions, then averaging). **The intra-rater Spearman correlation between holistic and rubric-based scoring is 0.57**—slightly lower than human-LLM agreement. This means the LLM scorer’s agreement with holistic human judgment ($r = 0.61$) is comparable to the agreement between two different human evaluation methods applied to the same material ($r = 0.57$). The absolute scores differ systematically (LLM mean 0.67, holistic human mean 0.61, rubric-based human mean 0.48), with the LLM being more generous and the

Table 7: Agentic feedback loop results (15 rejected relations).

Metric	Value
Relations evaluated	15
Score improved after revision	7/15 (47%)
Passed acceptance threshold (≥ 0.6)	8/15 (53%)
Content meaningfully changed	5/15 (33%)
Mean score before revision	0.39
Mean score after revision	0.52
Mean score change	+0.13
Closer to human judgment	3/15 (20%)

rubric approach being stricter, but the *relative ranking* produced by the LLM is no worse than cross-method human variability.

Two systematic LLM biases emerge. First, the LLM **over-rates creative but weakly-supported relations**: mapping abstract theories to specific metrics (e.g., Fama-French $\rightarrow$ Dividend Yield) receives inflated scores because the LLM finds the narrative pattern plausible even when the economic connection is tenuous. Second, the LLM **underrates tautological relations**: straightforward matches (e.g., Low PB Factor $\rightarrow$ Value Style Market) are scored conservatively because the LLM penalizes “obviousness.” The most concerning pattern is in the medium-quality range (LLM 0.5–0.8), where human-LLM agreement collapses to near-zero (Spearman $r = 0.02$)—the LLM cannot reliably discriminate quality within this band. This is precisely where the deliberative consensus protocol is designed to add value: when a single scorer model is uncertain or inconsistent, multi-scorer debate with convergence tracking provides a more robust assessment.

4.7 Feedback Loop Effectiveness

We evaluate the agentic feedback loop end-to-end on 15 relations that were rejected or scored weakly by the scorer agents (primary fused score < 0.6). For each, a scorer LLM (DeepSeek-V4-Pro) generates a structured critique identifying the specific issue (logic flaw, insufficient evidence, type mismatch, or overclaim), and the Generator LLM (same model, acting in generator role) revises the artifact based on the critique. The revised artifact is then re-scored.

Three patterns emerge. First, **major revisions are highly effective**: when the Generator makes substantive changes (e.g., revising “Low Beta Factor $\rightarrow$ High-Volatility Market” to “Low Beta Factor $\rightarrow$ Low-Volatility Market” after critique identified the original claim as directionally wrong), scores jump dramatically (0.38 $\rightarrow$ 0.95). Second, **the re-scorer correctly identifies unchanged garbage**: when the Generator refuses to revise a genuinely flawed relation (e.g., FF3 $\rightarrow$ PC), the re-score correctly drops (0.42 $\rightarrow$ 0.00), preventing passage of unimproved artifacts. Third, **the feedback loop improves scorer-judged quality but not necessarily human alignment** (only 20% closer to human): this

Table 8: API call budget: measured experiments and full-deployment projection.

Scenario	API Calls	
Measured experiments		
Deliberation (10 relations)	60	2 scorers
Feedback loop (15 relations)	45	critique + re-score
Heterogeneity study	182	5×30 primary scorers
Full-deployment projection (1,000 relations)		
Primary scoring (2 scorers)	2,000	
Deliberation (21.5% controversy)	860	2 scorers
Feedback loop (29.3% rejected)	188	critique + re-score
Total	3,048	
Single-pass (1 model, no verification)	1,000	
Cost per retained factor (CSI800, 111 cols)	27.5	
No-gate baseline (300+ cols retained)	3.3	

is consistent with the earlier finding that scorer and generator share similar LLM biases, and the feedback loop optimizes for scorer satisfaction rather than ground-truth quality. This divergence between scorer-endorsed improvement and human-judged improvement is a cautionary finding that underscores the importance of human validation in the verification loop.

The 53% pass rate after revision—with a mean score improvement of +0.13—demonstrates that the feedback loop can salvage a meaningful fraction of initially rejected artifacts. Combined with the deliberation protocol (80% convergence rate), this establishes a complete agentic quality-control pipeline: deliberation resolves scorer disagreement, and the feedback loop rehabilitates artifacts that fail even after consensus.

4.8 Deployment Cost Analysis

4.9 Deployment Cost Analysis

Multi-agent verification adds computational overhead proportional to the controversy rate among scorers. Table 8 summarizes the API call budgets from our experiments and extrapolates to a full-scale deployment processing 1,000 candidate relations.

The full multi-agent configuration requires approximately 3 $\times$ the API calls of a single-pass baseline (3,048 vs. 1,000). However, the cost per *retained high-quality factor* tells a different story: the full configuration spends 27.5 calls per retained factor (3,048 calls / 111 factors), while a no-gate baseline that retains 300+ noisy factors spends only 3.3 calls per retained factor but incurs substantial downstream costs in model training and portfolio execution on low-quality features. In practice, the deliberation and feedback loop overhead is concentrated on a minority of artifacts (21.5% trigger deliberation, 29.3% of those trigger feedback), meaning that 78.5% of artifacts incur only the 2 $\times$ primary scoring overhead. At current API pricing ($\sim$ \$1/M tokens for DeepSeek-V4-Pro), the full verification pipeline costs approximately

\$2.50–\$5.00 per 1,000 relations—a negligible fraction of the computational budget for downstream model training and backtesting.

5 Discussion

5.1 When Does Multi-Agent Verification Help?

The heterogeneous cross-market results establish a clear but nuanced pattern: multi-agent verification benefits CSI800, CSI1000, and CSI500 under the extended training window, while CSI300 is the exception. Two hypotheses merit consideration. First, **quality spectrum**: agent verification is most valuable when the generator produces a wide quality distribution—some excellent artifacts, some poor ones—and the scorers can discriminate between them. If the generator’s output quality is uniformly mediocre for a given market structure, verification cannot salvage unusable material. Second, **training horizon sensitivity**: the relative importance of verification gates shifts with the training window. CSC (earliest gate) dominates in the shorter-window setting, while EQ and SC (later gates) become more important with extended training, suggesting that different failure modes become rate-limiting as more data becomes available. The reversal of which gate matters most across windows is itself an informative finding for practitioners deploying multi-agent verification systems.

5.2 Architectural Heterogeneity vs. Homogeneity

Using different LLM backends for different scorer roles provides genuine evaluation diversity. We observe that GLM-5 and Kimi-K2.5 disagree most on *borderline* artifacts (fused score 0.5–0.7), where one model’s more conservative assessment counters the other’s leniency. This productive tension would be lost if a single model evaluated all dimensions. This finding is consistent with AI Council’s observation that homogeneous evaluator pools produce artificial consensus [1].

A practical implication is that model selection for scorer roles should prioritize *complementarity* over raw capability. Two medium-capability models that disagree productively may provide better verification than a single high-capability model that is consistently wrong in the same direction.

5.3 Limitations

Several boundaries should be stated. First, the evidence is for Chinese equities over 2022–2025; generalization to other markets (US, European, emerging), asset classes (bonds, commodities), and longer horizons requires further validation. Second, the scorer prompts, model choices, and thresholds ($\theta_d = 0.2$, $\theta_c = 0.1$, $\theta_a = 0.6$) are implementation-specific and were selected based on development-set analysis; different configurations may yield different results. Third, while our inter-agent communication protocol supports fully decentralized peer-to-peer communication, the current deployment uses a central orchestrator—a deliberate simplification

for reliability that should be relaxed in future work. Fourth, the portfolio evaluation, while including transaction costs, does not fully decompose industry, size, or liquidity effects in the primary specification.

5.4 Future Work

Three directions are natural next steps. (1) **Fully decentralized deployment**: removing the central orchestrator in favor of peer-to-peer agent communication, which would improve fault tolerance and enable truly distributed operation across cloud regions. (2) **Learned consensus weights**: replacing fixed scorer weights with learned weighting based on per-domain reliability, as demonstrated in A-HMAD. (3) **Cross-asset generalization**: deploying the same agent topology on different asset classes and geographies to map the boundary conditions more precisely.

6 Related Work

Our work sits at the intersection of three active research areas.

Multi-agent LLM verification. Recent work establishes that independent agent evaluation reduces self-verification bias through role separation and consensus mechanisms [1, 6, 10]. SC-MoA demonstrates that reading complete reasoning traces—rather than voting on final answers—recovers correct solutions even when all agents agree on a wrong answer, establishing that the unit of aggregation should be the reasoning trace, not the answer. A-HMAD introduces heterogeneous agent debate with dynamic role routing and learned consensus weighting, achieving 4–6% absolute gains on reasoning benchmarks through specialized agent roles. AI Council reveals a critical failure mode: homogeneous LLM evaluators artificially converge on the same option regardless of assigned perspectives, and architectural heterogeneity (different model architectures for different value perspectives) reduces first-choice concentration from 70.9% to 46.1%. Aegean reframes agent consensus as a distributed systems problem, introducing quorum-based early termination, stability windows that filter transient majorities, and streaming consensus that detects quorum during token generation—achieving 1.2–20× latency reduction and up to 4.4× token savings.

Our deliberative consensus protocol extends this line in three ways. First, while prior work applies consensus to final outputs, we apply staged verification gates at each intermediate semantic transition (relation, hypothesis, expression), preventing errors from propagating downstream. Second, our heterogeneity study empirically demonstrates the failure mode that prior work identifies theoretically: when models genuinely disagree, simple equal-weight averaging degrades performance in 69% of cases. Third, our integration of deliberation with an agentic feedback loop closes the loop from verification to revision, which prior work treats separately.

Distributed agent architectures. Google’s Agent-to-Agent (A2A) protocol standardizes agent discovery through

Agent Cards served at `/.well-known/agent.json` and task exchange through typed message passing, with support from 150+ organizations across major hyperscalers. Anthropic’s Model Context Protocol (MCP) complements A2A by standardizing agent-to-tool connectivity. Agent orchestration frameworks such as LangGraph, CrewAI, and AutoGen provide reusable patterns for multi-agent workflows including sequential pipelines, debate loops, and hierarchical delegation. Our inter-agent communication protocol adopts A2A-inspired design patterns—agent cards advertising capabilities and endpoints, typed message envelopes, structured artifact containers with provenance chains—while specializing message types (10 verification-specific types covering generation, scoring, deliberation, and revision) for staged quality control workflows that general-purpose frameworks do not natively support.

Automated factor discovery. Factor construction has progressed from hand-crafted formulas through evolutionary search [4], reinforcement learning [11], and symbolic generation [12] to LLM-assisted systems [7–9]. Knowledge-guided approaches constrain generation with domain structure [2, 3, 5]. Our work differs in treating the knowledge structure itself as an agent-constructed artifact requiring staged verification, rather than an externally trusted input. This perspective places our contribution at the intersection of multi-agent verification and automated factor discovery—a position that, to our knowledge, has not been explored in prior work.

7 Conclusion

We presented a distributed multi-agent verification architecture combining role-specialized scorer agents, a deliberative consensus protocol, and an agentic feedback loop. Deployed in a quantitative finance factor discovery case study across four Chinese equity markets (2022–2025, extended training window), the architecture demonstrates context-dependent effectiveness: it improves portfolio outcomes in three of four markets—CSI800 (ARR +0.091), CSI1000 (ARR +0.087), and CSI500 (ARR +0.099)—while CSI300 serves as a boundary condition. Agent ablation reveals that the relative importance of verification gates depends on the training horizon: CSC dominates in short-window settings while EQ and SC become rate-limiting with extended training. The deliberation protocol robustly resolves 80% of disagreements within 2 rounds (std reduction 0.155), and the agentic feedback loop successfully rehabilitates 53% of rejected artifacts (mean score improvement +0.13).

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